

AWS CLOUD SECURITY BASICS

Beginner-Level Cloud Security Learning Project

Prepared by	Anubhav Chaudhary
Program	BCA — Cloud & Cyber Security
University	IIMT University, Meerut
Project Type	Academic / Self-Learning Project
Technology Focus	Amazon Web Services (AWS) & Cloud Security

This project demonstrates foundational understanding of AWS cloud concepts and basic security practices. It is designed as a beginner project for an internship application and does not claim production AWS deployment experience.

1. Introduction

Cloud computing allows organizations to use computing resources such as servers, storage, databases, and networking services through the internet. AWS is a major cloud platform that provides a wide range of these services. Cloud security focuses on protecting cloud resources, identities, data, and applications from unauthorized access and other security risks.

The purpose of this project is to study basic AWS security concepts and understand how security practices can be applied when working with cloud resources.

2. Project Objectives

- Understand the basic concept of AWS and cloud computing.
- Understand the role of identity and access management in cloud security.
- Learn why strong authentication and least-privilege access are important.
- Understand basic security considerations for Amazon S3 storage.
- Create a simple conceptual AWS security architecture.
- Develop foundational knowledge relevant to a Cloud/Cyber Security internship.

3. AWS Services Covered

Service / Concept	Purpose in this project
AWS IAM	Identity and access management; controls who can access AWS resources.
Amazon S3	Cloud object storage; security includes access control and protection of stored data.
AWS MFA	Adds an additional authentication factor to help protect accounts.
CloudTrail	Relevant AWS service for recording API activity and supporting auditing.
VPC	Virtual networking environment used to organize and control network resources.

4. Identity & Access Management (IAM)

IAM is a central part of cloud security because permissions determine which identities can perform actions on AWS resources. A basic security approach is to provide users only the permissions required for their responsibilities instead of giving unnecessary access.

- Use separate identities rather than sharing credentials.
- Apply least-privilege permissions.
- Use strong authentication practices.
- Enable multi-factor authentication (MFA) where appropriate.
- Review permissions periodically and remove unnecessary access.

5. Amazon S3 Security Basics

Amazon S3 is an object storage service. Because cloud storage may contain sensitive information, access should be deliberately configured. Public access should not be enabled unless there is a specific and understood requirement.

- Keep sensitive buckets private by default.

- Use appropriate identity and resource-based permissions.
- Avoid storing credentials or secrets in objects.
- Use encryption options appropriate to the data and application.
- Monitor and review access to important storage resources.

6. Authentication & MFA

A password alone is a single authentication factor. Multi-factor authentication adds another factor, which can reduce the risk associated with compromised passwords. For cloud accounts with significant privileges, strong authentication practices are especially important.

7. Conceptual AWS Security Architecture

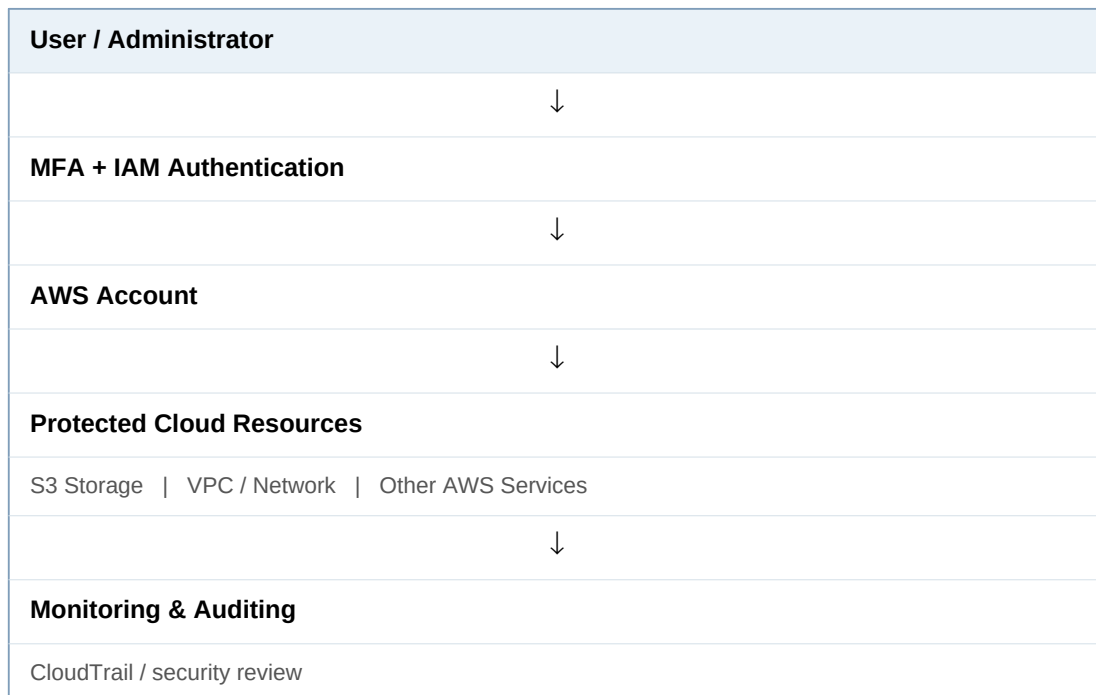


Figure 1: Conceptual security flow showing authentication, controlled access, protected resources, and monitoring. This is a learning architecture, not a record of a deployed production environment.

8. Methodology / Learning Process

- Reviewed foundational cloud computing and AWS concepts.
- Studied the purpose of IAM, MFA, S3 security, VPC concepts, and CloudTrail.
- Mapped common security practices to a simple conceptual AWS environment.
- Documented the security controls and their purpose in this report.
- Identified areas for future hands-on practice using AWS learning resources.

9. Key Findings

- Identity and permissions are fundamental to cloud security.
- Least privilege helps limit unnecessary access.
- MFA provides an additional authentication layer.
- Cloud storage requires careful access configuration.
- Logging and monitoring support investigation and auditing.

- Security should be considered throughout the design and operation of cloud resources.

10. Project Limitations

This is a beginner-level conceptual project. No production AWS environment, customer data, or real organizational infrastructure was used. The project demonstrates understanding of security concepts rather than professional deployment experience.

11. Future Scope

- Build a small AWS lab environment using IAM and S3.
- Practice creating least-privilege IAM policies.
- Configure secure S3 access and encryption in a controlled lab.
- Explore CloudTrail logs and basic security monitoring.
- Learn AWS networking and security services in greater depth.
- Document hands-on labs and publish future projects on GitHub.

12. Conclusion

This project provided a foundational understanding of AWS cloud security. It covered identity and access management, authentication, storage security, networking concepts, and auditing. The project also helped connect cloud computing concepts with practical security considerations and provides a foundation for further hands-on learning.

13. Relevant Training

Introduction to Cloud (AWS)

E & ICT Academy, IIT Kanpur

Training period: 01-09-2025 to 12-12-2025

Certificate issued: 03-08-2026

Declaration: This report was prepared as a beginner learning project to demonstrate foundational knowledge of AWS and cloud security. It should not be interpreted as evidence of professional AWS deployment experience.